

Krishnendu Chakrabarty

professor

基本信息

姓名: Krishnendu Chakrabarty

所属院校: 亚利桑那州立大学

地区: United States

年龄范围: 50岁以上

海外经历: 有

荣誉称号

★ AAAS Fellow

★ ACM Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

人工智能

集成电路设计与集成系统

信息安全

微电子科学与工程

电子与计算机工程

生物医学工程

研究领域

深度学习与神经网络

系统安全与漏洞分析

微电子器件与制造

EDA工具与电路验证

低功耗电路与能量管理

计算机体系结构

存储器设计与架构

超大规模集成电路设计

生物信号处理与模式识别

生物医学传感与检测技术

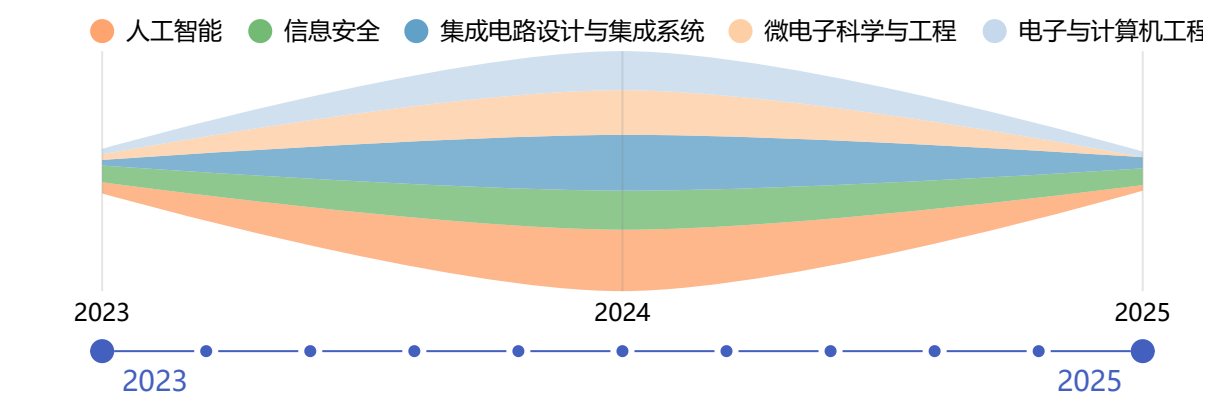
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个人简介

Krishnendu Chakrabarty is an Indian - American electrical and computer engineer. He holds a B. Tech from IIT Kharagpur (1990) and a Ph.D. from the

University of Michigan (1995). He's the Fulton Professor of Microelectronics at Arizona State University. Previously, he was at Duke and Boston Universities. He has received numerous awards, including the NSF Early Faculty Award, IEEE Circuits and Systems Society's Charles A. Desoer Technical Achievement Award, and the 2022 Semiconductor Research Corporation Aristotle Award. He's a fellow of ACM and AAAS, and a Senior Member of the National Academy of Inventors.

近年发表的论文



SPICED plus : Syntactical Bug Pattern Identification and Correction of Trojans in A/MS Circuits Using LLM-Enhanced Detection
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS
Chaudhuri, Jayeeta; Thapar, Dhruv; Chaudhuri, Arjun; Firouzi, Farshad; Chakrabarty, Krishnendu
被引用次数: 0 2025

TaintLock: Hardware IP Protection Against Oracle-Guided and Oracle-Reconstruction Attacks
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS
Talukdar, Jonti; Chaudhuri, Arjun; Ortega, Eduardo; Chakrabarty, Krishnendu
被引用次数: 0 2025

Detection of Voltage Droop-Induced Timing Fault Attacks Due to Hardware Trojans
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS
Talukdar, Jonti; Vyas, Akshay; Chakrabarty, Krishnendu
被引用次数: 0 2025

Discretized-Isolation Forest: Memory- and Compute-Efficient Unsupervised Anomaly Detection for Resource-Constrained Internet of Things Edge Devices
IEEE INTERNET OF THINGS JOURNAL
Ortega, Eduardo; Su, Fei; Chattopadhyay, Rita; Chakrabarty, Krishnendu

被引用次数: 0

2025

ML-TIME: ML-driven Timing Analysis of Integrated Circuits in the Presence of Process Variations and Aging Effects

2024 ACM/IEEE 6TH SYMPOSIUM ON MACHINE LEARNING FOR CAD, MLCAD 2024

Tan, Xuanyi; Domanski, Peter; Banerjee, Sanmitra; Chakrabarty, Krishnendu

被引用次数: 0

2024

Accelerating Fluid Loading in Sample Preparation with Fully Programmable Valve Arrays

PROCEEDINGS OF THE 37TH INTERNATIONAL CONFERENCE ON VLSI DESIGN, VLSID 2024 AND 23RD INTERNATIONAL CONFERENCE ON EMBEDDED SYSTEMS, ES 2024

Kumar, Mohit; Khan, Abhik Kumar; Roy, Sudip; Chakrabarty, Krishnendu; Bhattacharjee, Sukanta

被引用次数: 0

2024

Advancing blood glucose prediction with neural architecture search and deep reinforcement learning for type 1 diabetics

BIOCYBERNETICS AND BIOMEDICAL ENGINEERING

Domanski, Peter; Ray, Aritra; Lafata, Kyle; Firouzi, Farshad; Chakrabarty, Krishnendu; Pflueger, Dirk

被引用次数: 0

2024

BioTrojans: viscoelastic microvalve-based attacks in flow-based microfluidic biochips and their countermeasures

SCIENTIFIC REPORTS

Baban, Navajit Singh; Zhou, Jiarui; Elkhoury, Kamil; Bhattacharjee, Sukanta; Vijayavenkataraman, Sanjairaj; Gupta, Nikhil; Song, Yong-Ak; Chakrabarty, Krishnendu; Karri, Ramesh

被引用次数: 0

2024

Federated clustered multi-domain learning for health monitoring (vol 14, 903, 2024)

SCIENTIFIC REPORTS

Jiang, Shiyi; Li, Yuan; Firouzi, Farshad; Chakrabarty, Krishnendu

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Federated clustered multi-domain learning for health monitoring

SCIENTIFIC REPORTS

Jiang, Shiyi; Li, Yuan; Firouzi, Farshad; Chakrabarty, Krishnendu

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ML-TIME: ML-driven Timing Analysis of Integrated Circuits in the Presence of Process Variations and Aging Effects

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Tan, Xuanyi; Domanski, Peter; Banerjee, Sanmitra; Chakrabarty, Krishnendu

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Rowhammer Vulnerability of DRAMs in 3-D Integration

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ortega, Eduardo; Talukdar, Jonti; Paik, Woohyun; Bletsch, Tyler; Chakrabarty, Krishnendu

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PathDriver-Wash: A Path-Driven Wash Optimization Method for Continuous-Flow Lab-on-a-Chip Systems

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Huang, Xing; Wang, Jiaxuan; Yu, Zhiwen; Guo, Bin; Ho, Tsung-Yi; Schlichtmann, Ulf; Chakrabarty, Krishnendu

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Physically Unclonable Fingerprints for Authentication

APPLIED CRYPTOGRAPHY AND NETWORK SECURITY WORKSHOPS, PT II, ACNS 2024-AIBLOCK 2024, AIHWS 2024, AIOTS 2024, SCI 2024, AAC 2024, SIMLA 2024, LLE 2024, AND CIMSS 2024

Baban, Navajit S.; Zhou, Jiarui; Bhattacharya, Sarani; Chatterjee, Urbi; Bhattacharjee, Sukanta; Vijayavenkataraman, Sanjairaj; Song, Yong-Ak; Mukhopadhyay, Debdeep; Chakrabarty, Krishnendu; Karri, Ramesh

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2024

Detection of Stealthy Bitstreams in Cloud FPGAs using Graph Convolutional Networks

IEEE EUROPEAN TEST SYMPOSIUM, ETS 2024

Chaudhuri, Jayeeta; Chakrabarty, Krishnendu

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Test-Fleet Optimization Using Machine Learning

IEEE EUROPEAN TEST SYMPOSIUM, ETS 2024

Datta, Aniruddha; Yaganti, Bhanu Vikas; Dove, Andrew; Peltz, Arik; Chakrabarty, Krishnendu

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Testing and Fault Diagnosis for Multi-level Resistive Random-Access Memory in Monolithic 3D Integration

2024 IEEE 42ND VLSI TEST SYMPOSIUM, VTS 2024

Hung, Shao-Chun; Bhounik, Partho; Chakrabarty, Krishnendu

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Fault Diagnosis for Resistive Random Access Memory and Monolithic Inter-Tier Vias in Monolithic 3-D Integration

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Hung, Shao-Chun; Chaudhuri, Arjun; Banerjee, Sanmitra; Chakrabarty, Krishnendu

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Low-Overhead Clustered Federated Learning for Personalized Stress Monitoring

IEEE INTERNET OF THINGS JOURNAL

Jiang, Shiyi; Firouzi, Farshad; Chakrabarty, Krishnendu

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Block-Wise Mixed-Precision Quantization: Enabling High Efficiency for Practical ReRAM-Based DNN Accelerators

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Wu, Xueying; Hanson, Edward; Wang, Nansu; Zheng, Qilin; Yang, Xiaoxuan; Yang, Huanrui; Li, Shiyu; Cheng, Feng; Pande, Partha Pratim; Doppa, Janardhan Rao; Chakrabarty, Krishnendu; Li, Hai

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HuNT: Exploiting Heterogeneous PIM Devices to Design a 3-D Manycore Architecture for DNN Training

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DAWN: Efficient Trojan Detection in Analog Circuits Using Circuit Watermarking and Neural Twins

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Chaudhuri, Jayeeta; Bhattacharya, Mayukh; Chakrabarty, Krishnendu

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Mitigating Slow-to-Write Errors in Memristor-Mapped Graph Neural Networks Induced by Adversarial Attacks

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Control-Logic Synthesis of Fully Programmable Valve Array Using Reinforcement Learning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Huang, Xing; Cai, Huayang; Guo, Wenzhong; Liu, Genggeng; Ho, Tsung-Yi; Chakrabarty, Krishnendu; Schlichtmann, Ulf

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Inference Serving System for Stable Diffusion as a Service

2024 IEEE CLOUD SUMMIT, CLOUD SUMMIT 2024

Ray, Aritra; Dannull, Lukas; Firouzi, Farshad; Lafata, Kyle; Chakrabarty, Krishnendu

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Neuron grouping and mapping methods for 2D-mesh NoC-based DNN accelerators

JOURNAL OF PARALLEL AND DISTRIBUTED COMPUTING

Nacar, Furkan; Cakin, Alperen; Dilek, Selma; Tosun, Suleyman; Chakrabarty, Krishnendu

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Data Pruning-enabled High Performance and Reliable Graph Neural Network Training on ReRAM-based Processing-in-Memory Accelerators

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

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Root-Cause Analysis with Semi-Supervised Co-Training for Integrated Systems

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Analysis of Optical Loss and Crosstalk Noise in MZI-Based Coherent Photonic Neural Networks

JOURNAL OF LIGHTWAVE TECHNOLOGY

Shafiee, Amin; Banerjee, Sanmitra; Chakrabarty, Krishnendu; Pasricha, Sudeep; Nikdast, Mahdi

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Theoretical Patchability Quantification for IP-Level Hardware Patching Designs

29TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2024

Liu, Wei-Kai; Tan, Benjamin; Fung, Jason M.; Chakrabarty, Krishnendu

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ALT-Lock: Logic and Timing Ambiguity-Based IP Obfuscation Against Reverse Engineering

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Criticality Analysis of Ring Oscillators in FPGA Bitstreams

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Dynamic Task Remapping for Reliable CNN Training on ReRAM Crossbars

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Tung, Chung-Hsuan; Joardar, Biresh Kumar; Pande, Partha Pratim; Doppa, Janardhan Rao; Li, Hai (Helen); Chakrabarty, Krishnendu

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Attacking ReRAM-based Architectures using Repeated Writes

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Pruning Coherent Integrated Photonic Neural Networks

IEEE JOURNAL OF SELECTED TOPICS IN QUANTUM ELECTRONICS

Banerjee, Sanmitra; Nikdast, Mahdi; Pasricha, Sudeep; Chakrabarty, Krishnendu

Material-level countermeasures for securing microfluidic biochips

LAB ON A CHIP

Baban, Navajit Singh; Saha, Sohini; Jancheska, Sofija; Singh, Inderjeet; Khapli, Sachin; Khobdabayev, Maksat; Kim, Jongmin; Bhattacharjee, Sukanta; Song, Yong-Ak; Chakrabarty, Krishnendu; Karri, Ramesh

Transferable Graph Neural Network-Based Delay-Fault Localization for Monolithic 3-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Hung, Shao-Chun; Banerjee, Sanmitra; Chaudhuri, Arjun; Kim, Jinwoo; Lim, Sung Kyu; Chakrabarty, Krishnendu

Diagnosis of Malicious Bitstreams in Cloud Computing FPGAs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chaudhuri, Jayeeta; Chakrabarty, Krishnendu

Control-Logic Synthesis of Fully Programmable Valve Array Using Reinforcement Learning

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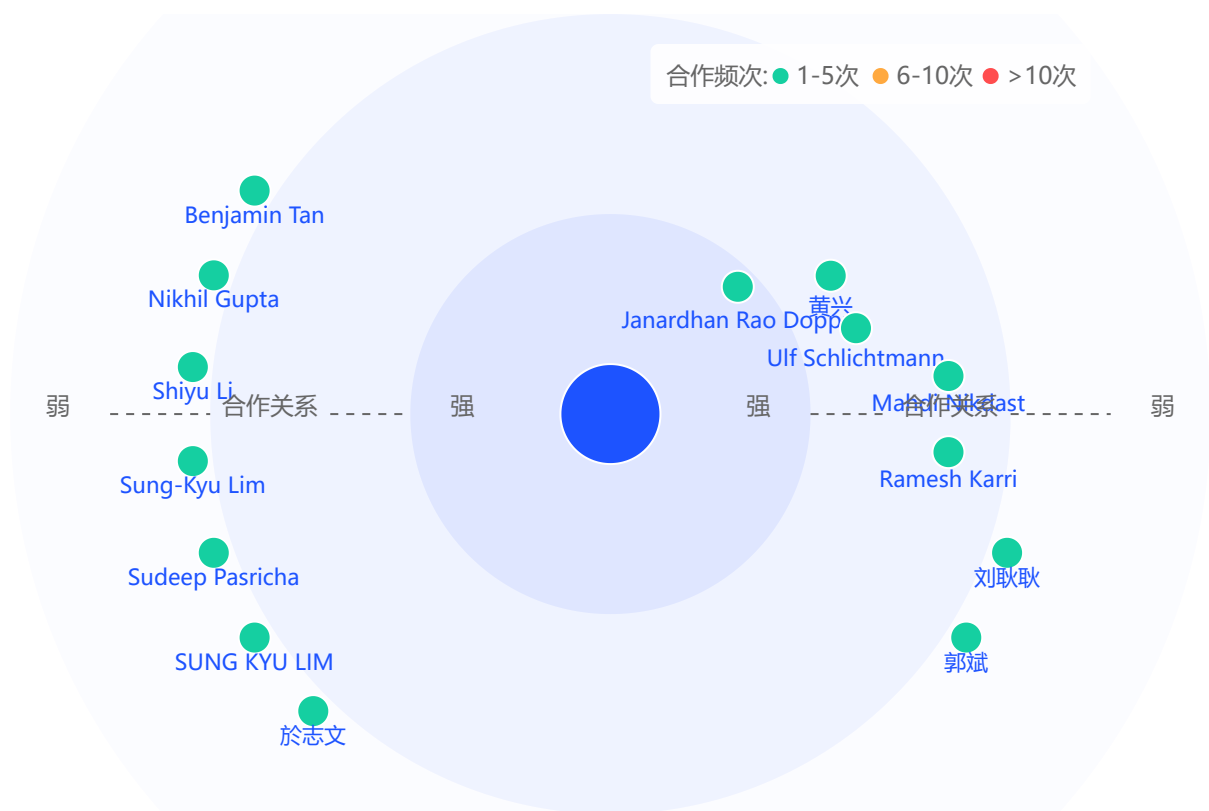
Xing Huang; Huayang Cai; Wenzhong Guo; Genggeng Liu; Tsung-Yi Ho; Krishnendu Chakrabarty; Ulf Schlichtmann

Transferable Graph Neural Network-Based Delay-Fault Localization for Monolithic 3-D ICs

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Shao-Chun Hung; Sanmitra Banerjee; Arjun Chaudhuri; Jinwoo Kim; Sung Kyu Lim; Krishnendu Chakrabarty

学术合作



Janardhan Rao Doppa

华盛顿州立大学-普尔曼

合作次数: 4

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